UNIVERSITY COLLEGE OF ENGINEERING

End Semester Examination

Discrete Structures & Theory of Logic (BCS303)

Roll No. _____

Subject: Discrete Structures & Theory of Logic

Time: 3 Hours

Max. Marks: 84

Code: BCS303

General Instructions:

1. Attempt all questions.
2. Show all necessary steps in mathematical derivations and proofs.
3. Hasse diagrams and graph representations must be clear and labeled.

SECTION A — SECTION A

Attempt all questions in this section.

2 marks each | Attempt all questions

Q.No.	Question	Marks	CO
1	Define a Modular Lattice and provide its necessary condition.	**2**	**CO1**
2	Define the Absorption Law as it applies to Boolean Algebra.	**2**	**CO2**
3	Define a Tautology in the context of Propositional Logic.	**2**	**CO3**
4	Write the contrapositive of the logical statement: 'If it rains, then the ground is wet'.	**2**	**CO3**
5	Define a Cyclic Group and state its generator property.	**2**	**CO4**
6	Define Isomorphism between two simple graphs.	**2**	**CO5**
7	Define a Bipartite Graph and state the condition for a graph to be bipartite.	**2**	**CO5**

SECTION B — SECTION B

Attempt all questions in this section.

7 marks each | Attempt all questions

Q.No.	Question	Marks	CO
8	Explain the properties of a Complemented Lattice and provide a Hasse diagram example of a lattice that is complemented but not distributed.	**7**	**CO1**
9	Simplify the Boolean function $F(A, B, C, D) = \Sigma(0, 1, 2, 5, 8, 9, 10)$ using a Karnaugh map and draw the resulting logic circuit.	**7**	**CO2**

Q.No.	Question	Marks	CO
10	Explain the Theory of Inference for Predicate Logic specifically focusing on the rules of Universal Generalization and Existential Specification.	**7**	**CO3**
11	State and prove Lagrange's Theorem regarding the order of subgroups in a finite group.	**7**	**CO4**
12	Compare Euler paths and Hamiltonian paths. Provide a graph that contains a Hamiltonian cycle but no Euler circuit.	**7**	**CO5**

SECTION C — SECTION C

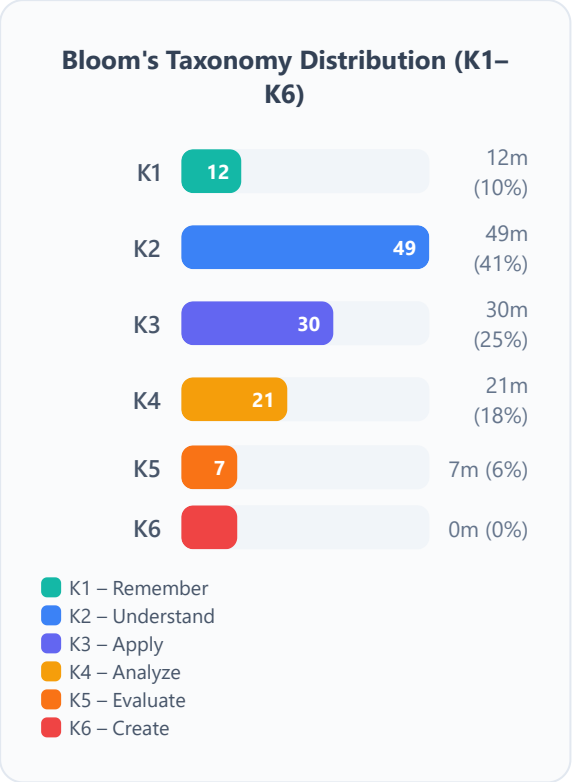
Attempt one part (a or b) from each question.

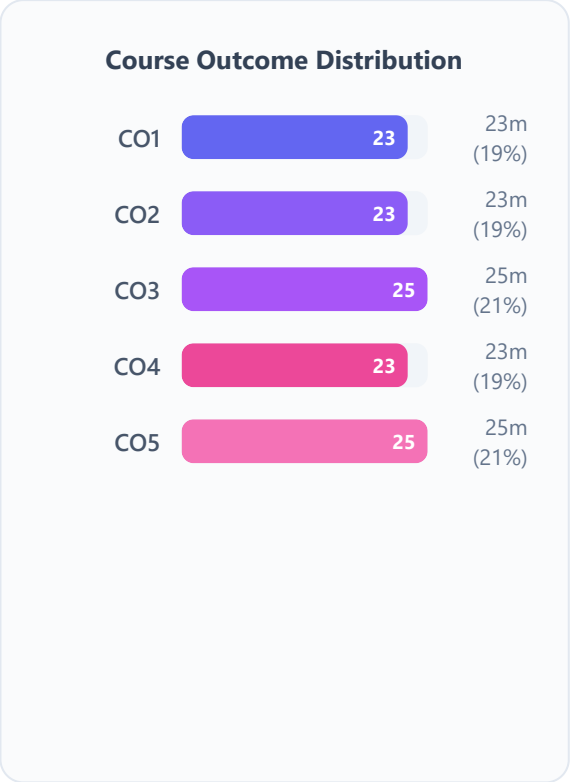
7 marks each | Attempt all questions

Q.No.	Question	Marks	CO
13	**(a)** Explain the recursive definition of relations and define the composition of two relations with a numerical example.	**7**	**CO1**
	OR		
	(b) Construct the Hasse diagram for the 'divides' relation on the set A = {1, 2, 3, 4, 6, 12} and determine if the resulting POSET is a lattice.	**7**	**CO1**
14	**(a)** Classify functions based on their properties (Injection, Surjection, and Bijection) and provide mathematical proofs for each classification using specific functions.	**7**	**CO2**
	OR		
	(b) Explain the growth of functions using Big-Oh (O) and Big-Omega (Ω) notations. Formally prove that f(n) = 3n^2 + 4n + 5 is O(n^2).	**7**	**CO2**
15	**(a)** Analyse the validity of the following argument using truth tables: 'If I study hard, I will pass the exam. I did not pass the exam. Therefore, I did not study hard.'	**7**	**CO3**
	OR		
	(b) Explain the procedure to convert a logical formula into its Prenex Normal Form (PNF) and provide a step-by-step example.	**7**	**CO3**
16	**(a)** Define a Ring and explain the elementary properties of Fields. Contrast a Ring with a Field using their respective axioms.	**7**	**CO4**
	OR		
	(b) Explain the concepts of Normal Subgroups and Group Homomorphisms. Prove that the kernel of a homomorphism is a normal subgroup.	**7**	**CO4**
17	**(a)** Explain the concept of Planar graphs and prove Euler's formula (V - E + F = 2) for any connected planar graph.	**7**	**CO5**
	OR		

Q.No.	Question	Marks	CO
	(b) Explain the principles of graph coloring and determine the chromatic number of a Peterson graph, justifying your answer.	**7**	**CO5**

Question Paper Analysis





Bloom's Taxonomy Summary		
Level	**Marks**	**%**
K1	12	10%
K2	49	41%
K3	30	25%
K4	21	18%
K5	7	6%
K6	0	0%

Course Outcome Summary		
CO	**Marks**	**%**
CO1	23	19%
CO2	23	19%
CO3	25	21%
CO4	23	19%
CO5	25	21%